

ADITYA GIRISH

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SUMMARY

Machine Learning Engineer with experience in LLMs, RAG systems, and deep learning. Published researcher with production ML deployments and applied experience in financial data modelling.

EDUCATION

Bachelor of Technology (B.Tech.), Computer Science, 2022-2026 **PES University, Bengaluru**
Honors & Scholarships: MRD Scholarship - Top 5% ; CNR Scholarship — two-time recipient, Top 10%.

PUBLICATIONS

MSTRAE: Multi-Scale Temporal Recurrent Autoencoder for Flight Manoeuvre Recognition
Future Technologies Conference (FTC) 2026 **Accepted — Oct 2026, Berlin**
Springer Lecture Notes in Networks and Systems · Indexed: Web of Science, SCOPUS, INSPEC *Lead Author*
Trained a multi-scale LSTM autoencoder in PyTorch with HMM-based unsupervised clustering and an LLM-based evaluation framework for zero-shot flight manoeuvre recognition on Cessna 172 telemetry data.

SKILLS

Languages	Python, SQL, Java, JavaScript, R
Frameworks	PyTorch, HuggingFace Transformers, Scikit-Learn, Pandas, NumPy, SpaCy, NLTK
Cloud & Tools	GCP, AWS, Docker, Git, Snowflake, Databricks
ML & AI	Machine Learning, Deep Learning, NLP, LLMs, RAG, Time Series Analysis
Methods	Statistical Modeling, Feature Engineering, Model Evaluation, Model Deployment, Fine-Tuning
Data Science	Data Analysis, Data Pipelines, Exploratory Data Analysis, A/B Testing, Hypothesis Testing
Architecture	Vector Databases, Transformer Models, LSTM, Autoencoders, Clustering, Classification
Coursework	Financial Markets (Yale), Statistics 110x (HarvardX), LLMs & Applications (TA)

EXPERIENCE

Software Engineering Intern – AI **Jan 2026 – Present**
TruEstate *Bengaluru, India*

- Built a RAG-based system using vector retrieval and LLM reasoning to automate customer communication, reducing manual intervention by **45%**.
- Designed a multi-agent knowledge base pipeline using ChromaDB, automating document ingestion, chunking, and embedding across multiple data sources.
- Deployed an end-to-end ML pipeline on GCP, increasing customer engagement by **35%** and reducing acquisition costs by **40%**.

AI Backend Engineer Intern **Apr 2025 – May 2025**
Viable Ideas Private Limited *Bengaluru, India*

- Trained an LLM-based inventory prediction model on financial transaction data from **4 bank sources**, achieving **85% classification accuracy** in production.
- Implemented automated data pipelines for multi-source financial data, reducing manual workflows by **60%** and operational costs by **40%**.

PROJECTS

QRAG : Quantized Bi-Level Retrieval Augmented Generation
• Developed a memory-efficient RAG system using dual-level quantization, achieving **3-4x memory reduction** and **95.1% recall@10** on the FIQA dataset.
• Benchmarked against FAISS baselines across 57,000 documents; outperformed IVF-PQ by **21.5pp in recall** at comparable memory budgets.

FinFusion : Finance Risk Fusion Model
• Constructed a deep learning model using LSTM forecasting and sentiment analysis for stock prediction, achieving **9.78% MAPE** and **68.9% directional precision**.
• Applied multi-input fusion of time-series and NLP features for buy/hold/sell classification across equity datasets.

Equilibrium : Autonomous Multi-Agent Supply Chain Automation (Invited Contributor, Aden — YC W20)
• Contributed to a multi-agent system using the Hive framework to automate supply chain contract workflows with real-time negotiation and verification.
• Established workflow logic and state management for procurement automation, ensuring reliability across agent coordination.